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Minimax Goodness-of-fit testing in High-Dimensional models

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Chapter 1

Introduction

Chapter 2

Minimax optimal testing by classification for Gaussian models

2.1 Setting of the problem

Definition 1. Let μ, ν be two probability measures on a space $\mathcal{X}$, we define the total variation between μ and ν as follows :

$$TV(\mu, \nu) = \sup_{E \subset \mathcal{X} \text{ measurable}} |P(E) - Q(E)|$$

If μ, ν have densities f, g respectively with respect to another measure λ , then one has the equality :

$$TV(\mu, \nu) = \int |p(x) - q(x)| d\lambda(x)$$

Now, suppose that one has two samples X, Y of size n from two distributions $\mathbb{P}_X$ and $\mathbb{P}_Y$ respectively on some space $\mathcal{X}$ and wishes to test the hypotheses :

$$\mathcal{H}_0 : \mathbb{P}_X = \mathbb{P}_Y \text{ versus } \mathcal{H}_1 : TV(\mathbb{P}_X, \mathbb{P}_Y) \geq \varepsilon$$

for some $\varepsilon > 0$, by using machine learning classifiers. Given such hypotheses $\mathcal{H}_0, \mathcal{H}_1$ seen as families of distributions on $\mathcal{X}$, we say that a decision rule $\psi : \mathcal{X} \rightarrow \{0, 1\}$ tests the two hypothesis against each other with error at most $\delta \in \left\{0, \frac{1}{2}\right\}$ if :

$$\max_{i=0,1} \max_{P \in \mathcal{H}_i} \mathbb{P}_{S \sim P}(\psi(S) \neq i) \leq \delta$$

The idea of using classifiers, firstly described by Friedman [2004], consists of training a binary classifier $\mathcal{C} : \mathcal{X} \rightarrow \{0, 1\}$ on the labeled data $\cup_{i=1}^n \{(X_i, 0), (Y_i, 0)\}$ and compare the samples $\mathcal{C}(X_1), \dots, \mathcal{C}(X_n)$ to $\mathcal{C}(Y_1), \dots, \mathcal{C}(Y_n)$. **ADD MORE DETAILS**

We will follow the framework described by Gerber et al. [2023] as "Classifier-Accuracy testing" (CAT) for goodness-of-fit testing :

Suppose we have a sample X of size $2n$ from an unknown distribution $\mathbb{P}_X$ that we will split into 2 halves $X = X^{\text{tr}} \sqcup X^{\text{te}}$ and another sample Y of size $2n$ from a known distribution $\mathbb{P}_Y$ which corresponds to our null hypothesis, we will split this sample in the same manner as for the previous sample into $Y = Y^{\text{tr}} \sqcup Y^{\text{te}}$. We train a classifier $\mathcal{C} : \mathcal{X} \rightarrow \{0, 1\}$ on the input $\mathcal{D}^{\text{tr}} := \{X^{\text{te}}, Y^{\text{te}}\}$ that aims to assign 1 to $\mathbb{P}_X$ and 0 to $\mathbb{P}_Y$. In practice, it is easier to think of $\mathcal{C}$ in terms of the separating set $S := \mathcal{C}^{-1}(\{1\})$ which is a random set whose randomness come from the splitting of our samples and potentially an external seed.

Definition 2. Given two datasets $\{A_i\}_{i=1}^a, \{B_j\}_{j=1}^b$, we define the classifier-accuracy statistic s :

$$T_S(A, B) := \frac{1}{a} \sum_{i=1}^a \mathbb{1}_{A_i \in S} - \frac{1}{b} \sum_{j=1}^b \mathbb{1}_{B_j \in S}$$

Finally, we threshold the output $|T_S|$ of the test data $\mathcal{D}^{\text{te}} := \{X^{\text{te}}, Y^{\text{te}}\}$

2.1.1 The Sobolev ellipsoid and Gaussian sequence model

In this project, we will restrict ourselves to a very specific class of distributions, the Gaussian sequence model with mean vector in the Sobolev ellipsoid that we will define below.

Definition 3. Let $s > 0, C > 0$. We define the Sobolev ellipsoid $\mathcal{E}(s, C)$ as follows :

$$\mathcal{E}(s, C) := \left\{ \theta \in \mathbb{R}^{\mathbb{N}} : \sum_{j=1}^{\infty} j^{2s} \theta_j^2 \leq C \right\}$$

For a sequence $\theta \in \mathcal{E}(s, C)$, we define $\mu_{\theta} = \bigotimes_{j=1}^{\infty} \mathcal{N}(\theta_j, 1)$. The set of Gaussian sequence model distributions is hence defined as :

$$\mathcal{P}_G(s, C_G) := \{ \mu_{\theta} ; \theta \in \mathcal{E}(s, C_G) \}$$

ADD MORE DETAILS ABOUT THE INTUITION BEHIND GAUSSIAN SEQUENCE MODEL

2.2 Preliminary results :

The hard part of this kind of frameworks consists in finiding a good separating set $S \subset \mathcal{X}$, but we firstly need to define what "good" means.

Definition 4. Let $S \subseteq \mathcal{X}$ be a measurable set, and μ, ν be two probability measures on $\mathcal{X}$. We define the separation and size of S respectively as follows :

$$\text{sep}(S) = \mu(S) - \nu(S) \text{ and } \tau(S) = \min(\mu(S)\mu(S^c), \nu(S)\nu(S^c))$$

Lemma 1. Gerber et al. [2023, Lemma 1]

Consider the hypothesis testing problem $\mathcal{H}_0 : p = q$ versus any other alternative $\mathcal{H}_1$. Suppose that the learner has constructed a separating set S such that :

$$\forall \mu, \nu \in \mathcal{H}_1 : |\text{sep}(S)| = |\mu(S) - \nu(S)| \geq \underline{\text{sep}} \text{ and } \forall \mu, \nu \in \mathcal{H}_0 \cup \mathcal{H}_1 : \min(\mu(S)\mu(S^c), \nu(S)\nu(S^c)) \leq \bar{\tau}$$

Then, using only the knowledge of $\bar{\tau}$, the classifier-accuracy test with samples of size n from both p and q and an appropriate threshold achieves type-I and type-II errors at most δ provided that :

$$n \geq c \frac{\log(\frac{1}{\delta})}{\underline{\text{sep}}} \left(1 + \frac{\bar{\tau}}{\underline{\text{sep}}} \right)$$

for a large enough constant $c > 0$

Remark 1. With Lemma 1, one can imagine that a separating can be referred to as "good" if $|\text{sep}(S)|$ is large under H_1 , and $\tau(S)$ is small under both H_0 and $\mathcal{H}_1$ with probability $1 - \delta$

In order to prove lemma 1, we will need a prior result on the Bernstein concentration of the classifier accuracy statistics.

Lemma 2. Gerber et al. [2023, Lemma 8] Suppose $A_1, A_2, \dots, A_n \stackrel{iid}{\sim} \text{Ber}(p)$ and $B_1, B_2, \dots, B_m \stackrel{iid}{\sim} \text{Ber}(q)$. Let $\tau = \min(p(1-p), q(1-q))$, and let $\bar{A}, \bar{B}$ be their empirical means, respectively. There exists a universal constant $c > 0$ such that :

$$\begin{aligned} \mathbb{P} \left(|\bar{A} - \bar{B}| \leq \frac{1}{2} |p - q| - \frac{1}{2} \sqrt{\frac{c \log(1/\delta) \tau}{\min(n, m)}} - \frac{1}{2} \frac{c \log(1/\delta)}{\min(n, m)} \right) &\leq \delta, \\ \mathbb{P} \left(|\bar{A} - \bar{B}| \geq 2 |p - q| + 2 \sqrt{\frac{c \log(1/\delta) \tau}{\min(n, m)}} + 2 \frac{c \log(1/\delta)}{\min(n, m)} \right) &\leq \delta \end{aligned}$$

Proof of lemma 1. Using n test samples from (X, Y) , consider the following classifier-accuracy test: we accept $\mathcal{H}_0$ if

$$\left| \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{X_i \in S} - \mathbb{1}_{Y_i \in S} \right| \leq \sqrt{\frac{c \bar{\tau} \log(1/\delta)}{n}} + \frac{c \log(1/\delta)}{n}$$

and reject the null hypothesis otherwise. Above, $c > 0$ is a large absolut constant, and note that the threshold only relies on the knowledge of $\bar{\tau}, n$ and δ $\square$

Chapter 3

Optimal transport based methods in Goodness-of-Fit problems

Chapter 4

Conclusion

Bibliography

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